

Light & Optics

Light & its Nature

DEFINITION

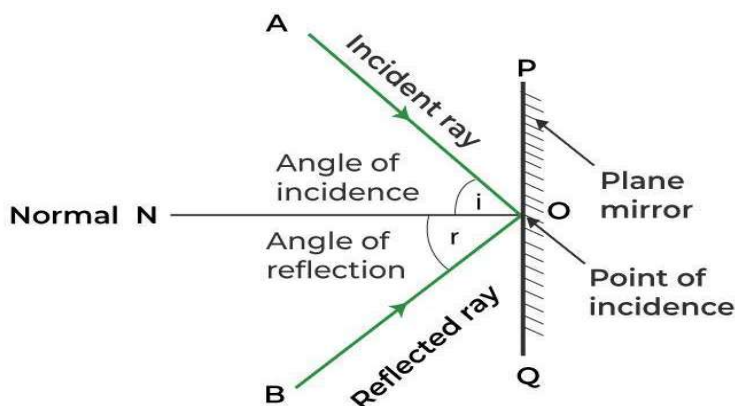
Light: A form of electromagnetic radiation that is visible to the human eye and enables us to see objects.

- **Rectilinear propagation:** Light travels in straight lines — proved by the fact that a small source of light casts a sharp shadow of an opaque object.
- **Speed of light in vacuum:** $c = 3 \times 10^8 \text{ m s}^{-1}$
- When an opaque object on the path of light is very small, light tends to bend around it — this effect is called **diffraction**. In such cases the straight-line (ray) treatment fails and the wave model must be applied.
- The wave theory of light is sometimes inadequate for describing the interaction of light with matter; light then behaves like a stream of particles. Modern quantum theory of light reconciles both properties.

Reflection of Light

DEFINITION

Reflection of Light: The process by which a ray of light bounces back into the same medium when it strikes a polished or smooth surface.



Laws of Reflection

Law 1: The angle of incidence ($\angle i$) is always equal to the angle of reflection ($\angle r$).

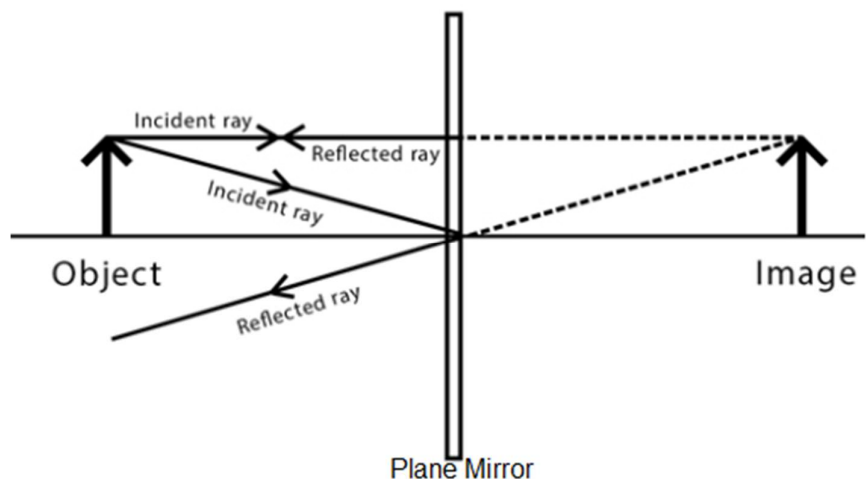
Law 2: The incident ray, the normal to the mirror at the point of incidence, and the reflected ray — all lie in the same plane.

★ Remember

Laws of reflection apply to ALL types of reflecting surfaces, including plane mirrors and spherical mirrors.

Properties of Image Formed by a Plane Mirror

- **Virtual and Erect**
- **Same size** as the object
- The image is as far **behind** the mirror as the object is **in front**
- **Laterally inverted** — the left and right sides are interchanged



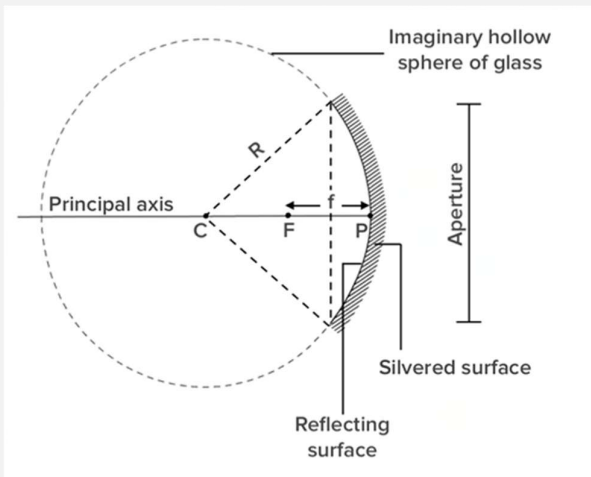
Spherical Mirrors

DEFINITION

Spherical Mirror: A mirror whose reflecting surface forms a part of a sphere.

Concave Mirror

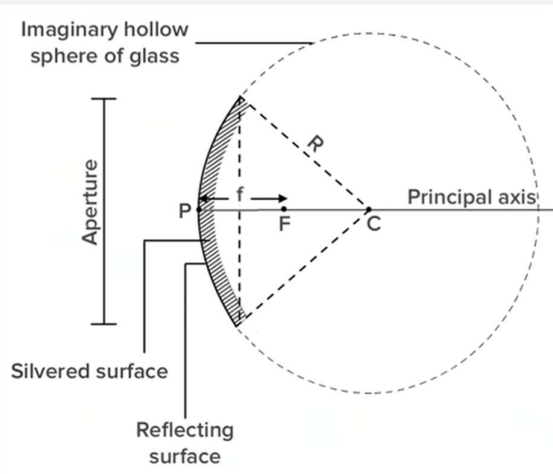
- Reflecting surface is curved inward (faces towards the centre of the sphere)



- Also called a converging mirror
- Centre of curvature lies in front of the mirror
- Used in: torches, headlights, shaving/make-up mirrors, solar furnaces, dentist mirrors

Convex Mirror

- Reflecting surface is curved outward (faces away from the centre of the sphere)



- Also called a diverging mirror
- Centre of curvature lies behind the mirror
- Used in: rear-view (wing) mirrors in vehicles

Important Terms and Definitions

- Pole (P):** The centre of the reflecting surface of a spherical mirror. It lies on the surface of the mirror.
- Centre of Curvature (C):** The centre of the sphere of which the reflecting surface of the mirror forms a part. It lies outside the reflecting surface.
- Radius of Curvature (R):** The radius of the sphere of which the mirror is a part. It equals the distance PC.
- Principal Axis:** The imaginary straight line passing through the pole P and the centre of curvature C. It is normal to the mirror at its pole.
- Principal Focus (F):** The point on the principal axis where rays parallel to the principal axis converge after reflection (concave mirror) or appear to diverge from (convex mirror) after reflection.
- Focal Length (f):** The distance between the pole P and the principal focus F.
- Aperture:** The effective diameter (MN) of the circular reflecting surface of a spherical mirror.

Relation between Focal Length and Radius of Curvature

$$R = 2f$$

where R is Radius of curvature

F is focal length.

★ Remember

The principal focus F lies **midway between the pole P and the centre of curvature C** for spherical mirrors of small aperture.

Focal Length of plane mirror is always infinity.

Rules for Drawing Ray Diagrams – Spherical Mirrors

- **Rule 1:** A ray parallel to the principal axis, after reflection from a concave mirror, passes through the principal focus F. For a convex mirror, the reflected ray appears to diverge from F.
- **Rule 2:** A ray passing through the principal focus F of a concave mirror (or directed towards F for a convex mirror), after reflection, emerges parallel to the principal axis.
- **Rule 3:** A ray passing through the centre of curvature C of a concave mirror (or directed towards C for a convex mirror), after reflection, is reflected back along the same path.
- **Rule 4:** A ray incident obliquely to the principal axis, towards the pole P of the mirror, is reflected obliquely — the incident and reflected rays make equal angles with the principal axis at P.

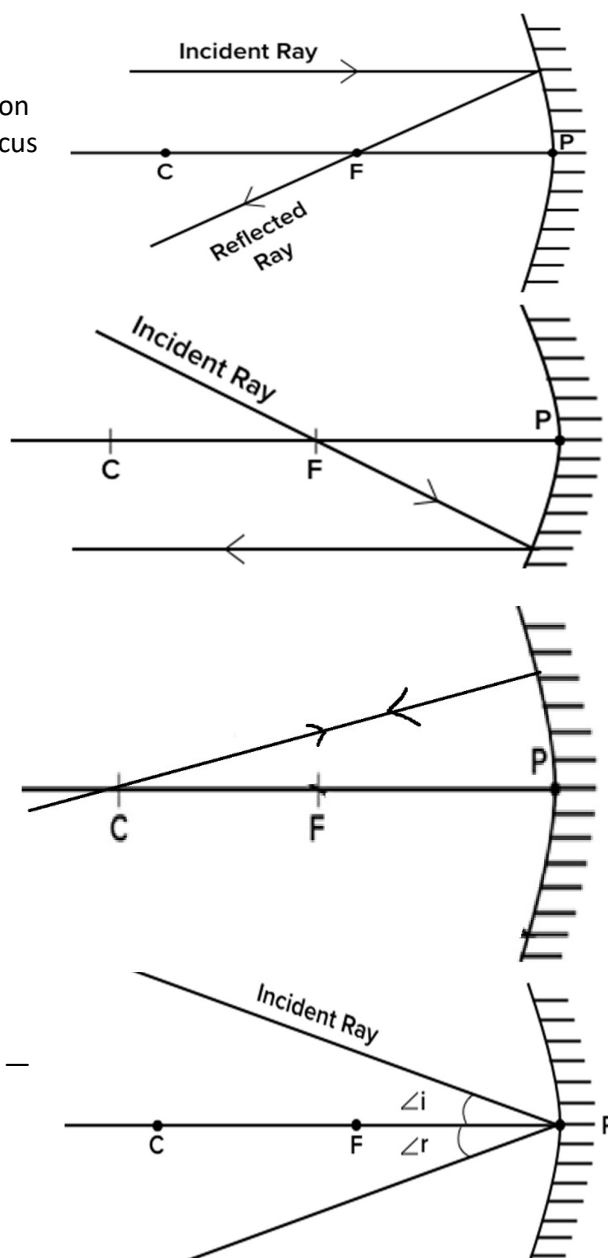
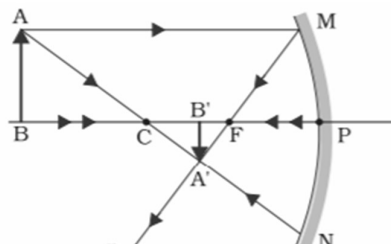
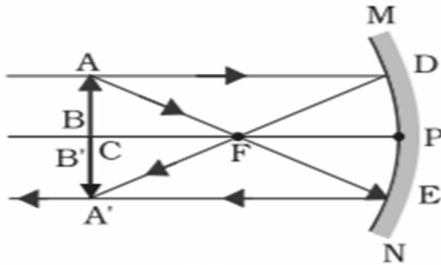
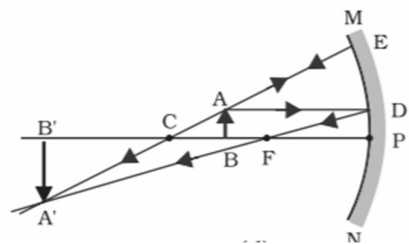


Image Formation by Concave Mirror

Case 1 : Object at Infinity	
Ray diagram	
Object Position	At Infinity — object is placed infinitely far from the mirror
Object Distance (u)	$u \rightarrow -\infty$ (tends to negative infinity)
Image Location	At the Principal Focus F — in front of the mirror, on the principal axis
Nature of Image	Real and Inverted
Size of Image	Highly diminished — almost a point (much smaller than the object)
Magnification (m)	$m \rightarrow 0$ (tends to zero from the negative side)

Case 2: Object Beyond C	
Ray diagram	
Object Position	Beyond C — object is placed farther from the mirror than the centre of curvature
Object Distance (u)	$u < -2f$ (magnitude greater than $2f$)
Image Location	Between F and C, in front of the mirror
Nature of Image	Real and Inverted
Size of Image	Diminished — smaller than the object
Magnification (m)	$0 < m < 1$ (m is negative; magnitude less than 1)
Case 3: Object at C	
Ray diagram	
Object Position	At C — object is placed exactly at the centre of curvature
Object Distance (u)	$u = -2f$ (magnitude equals $2f$)
Image Location	At C — image forms at the exact same position as the object, in front of mirror.
Nature of Image	Real and Inverted
Size of Image	Same size as the object — neither magnified nor diminished
Magnification (m)	$m = -1$ (magnitude exactly 1, negative sign means inverted)
Case 4 Object Between C and F	
Ray diagram	
Object Position	Between C and F — object is placed between the centre of curvature and focus.
Object Distance (u)	$-2f < u < -f$ (magnitude between f and $2f$)
Image Location	Beyond C — in front of the mirror.
Nature of Image	Real and Inverted
Size of Image	Enlarged — larger than the object
Magnification (m)	$ m > 1$ (m is negative; magnitude greater than 1)

Case 5: Object at F	
Ray diagram	
Object Position	At F — object is placed exactly at the principal focus of the mirror
Object Distance (u)	$u = -f$ (magnitude equals f)
Image Location	At Infinity — reflected rays emerge parallel and never converge to form an image
Nature of Image	Real and Inverted (at infinity — not practically obtainable on a screen)
Size of Image	Infinitely large — size tends to infinity
Magnification (m)	$m \rightarrow -\infty$ (tends to negative infinity)
Case 6: Object Between P and F	
Ray diagram	
Object Position	Between P and F — object is placed between the pole and the principal focus (very close to the mirror)
Object Distance (u)	$0 > u > -f$ (magnitude less than f)
Image Location	Behind the mirror — virtual image formed on the same side as the object
Nature of Image	Virtual and Erect (cannot be caught on a screen)
Size of Image	Enlarged — larger than the object
Magnification (m)	$m > +1$ (positive = erect; magnitude greater than 1 = enlarged)

Summary Table – Image by Concave Mirror

Position of Object	Position of Image	Size of Image	Nature
At infinity	At focus F	Highly diminished, point-sized	Real, Inverted
Beyond C	Between F and C	Diminished	Real, Inverted
At C	At C	Same size	Real, Inverted
Between C and F	Beyond C	Enlarged	Real, Inverted
At F	At infinity	—	Image not formed
Between P and F	Behind mirror	Enlarged	Virtual, Erect

Image Formation by Convex Mirror

Case 1: Object at Infinity	
Ray diagram	
Object Position	At Infinity — object is placed infinitely far from the convex mirror
Object Distance (u)	$u \rightarrow -\infty$ (tends to negative infinity)
Image Location	At the virtual Principal Focus F — behind the mirror, on the principal axis
Nature of Image	Virtual and Erect (cannot be caught on a screen)
Size of Image	Highly diminished — almost a point
Magnification (m)	$m \rightarrow 0+$ (positive, tends to zero from above)
Case 2: Object at Any Finite Distance	
Ray diagram	
Object Position	Between infinity and P — object placed at any finite distance from the convex mirror
Object Distance (u)	Any value: $0 > u > -\infty$ (any negative value)
Image Location	Between P and F — behind the mirror; image moves from F toward P as object moves closer
Nature of Image	Virtual and Erect (cannot be caught on a screen)
Size of Image	Diminished — always smaller than the object
Magnification (m)	$0 < m < 1$ (positive = erect; magnitude less than 1 = diminished)

Summary Table – Image by Convex Mirror

Position of Object	Position of Image	Size	Nature
At infinity	At focus F , behind mirror	Highly diminished, point-sized	Virtual, Erect
Between ∞ and P	Between P and F , behind	Diminished	Virtual, Erect
★ Remember Convex mirrors always produce virtual, erect, and diminished images. That is why they are preferred as rear-view mirrors — they give an erect image and cover a wider field of view.			

New Cartesian Sign Convention

New Cartesian Sign Convention for spherical mirrors: origin at pole P, right is positive, left is negative, up is positive, down is negative.

NEW CARTESIAN SIGN CONVENTION

- The **pole (P)** is taken as the origin and the **principal axis** as the x-axis.
- The object is always placed to the **left** of the mirror, so the incident light always comes from the left.
- All distances measured to the **right of P** (along direction of incident light): **Positive (+ve)**
- All distances measured to the **left of P** (against direction of incident light): **Negative (-ve)**
- Heights **above** the principal axis: Positive (+ve)
- Heights **below** the principal axis: Negative (-ve)

Mirror Formula and Magnification

MIRROR FORMULA

Relates the object distance (u), image distance (v), and focal length (f) of a spherical mirror.

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

where: **v** = image distance, **u** = object distance, **f** = focal length (all measured from pole P using sign convention).

MAGNIFICATION

Magnification (m): The ratio of the height of the image to the height of the object. It gives the relative extent to which the image is magnified or diminished compared to the object.

$$m = \frac{h_i}{h_o} = \frac{-v}{u}$$

Where **h** is height of image and **h_o** is height of object.

INTERPRETING MAGNIFICATION

- **m is negative** → Image is **Real and Inverted**
- **m is positive** → Image is **Virtual and Erect**
- **|m| > 1** → Image is Enlarged; **|m| < 1** → Diminished; **|m| = 1** → Same size as object

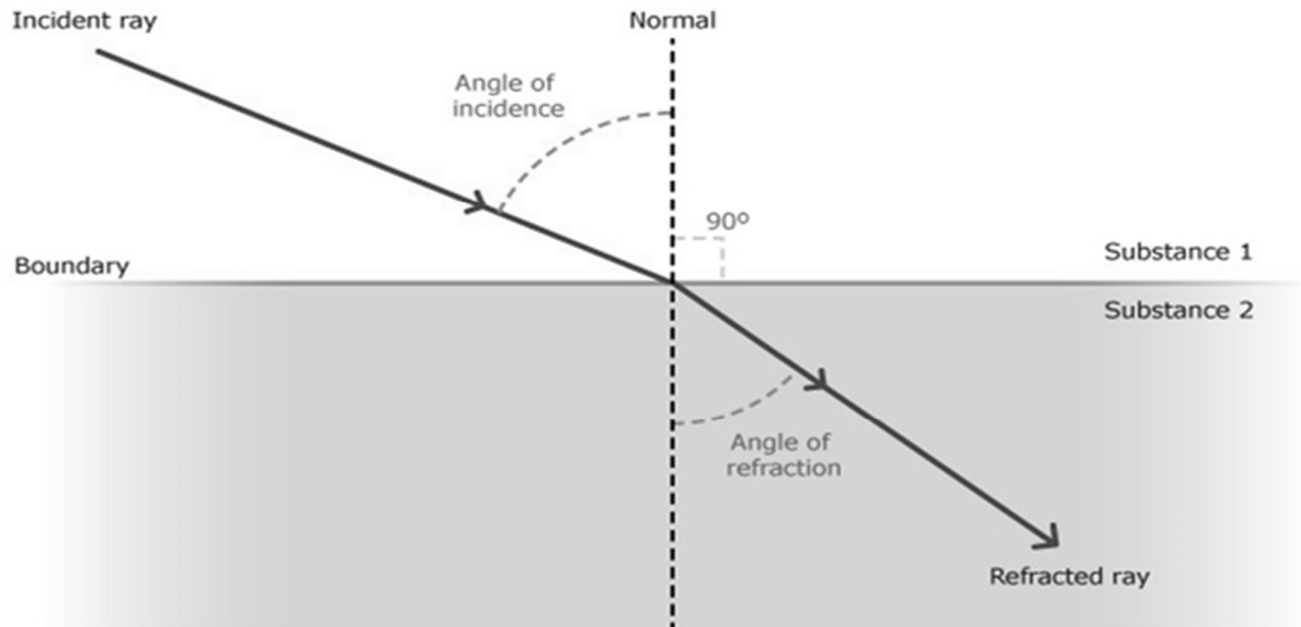
Refraction of Light

DEFINITION

Refraction of Light: The phenomenon of bending of light when it passes obliquely from one transparent medium to another. It occurs due to the change in speed of light as it moves from one medium to another.

Laws of Refraction (Snell's Law)

- **Law 1:** The incident ray, the refracted ray, and the normal to the interface of the two media at the point of incidence — all lie in the **same plane**.
- **Law 2 (Snell's Law):** The ratio of the sine of the angle of incidence to the sine of the angle of refraction is a constant for the light of a given colour and for the given pair of media. This constant is called the refractive index.



Snell's Law

$$\frac{\sin i}{\sin r} = \text{Constant} = \text{Refractive Index}(n_{21})$$

Rarer → Denser Medium

- Speed of light decreases
- Light bends TOWARDS the normal
- $\angle r < \angle i$ (angle of refraction is smaller)

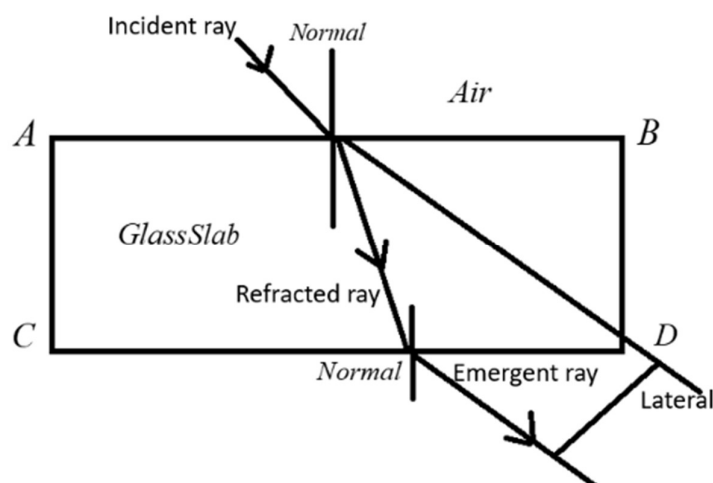
Denser → Rarer Medium

- Speed of light increases
- Light bends AWAY from the normal
- $\angle r > \angle i$ (angle of refraction is larger)

Examples of refraction in Practical Life

- The bottom of a tank or pond containing water appears to be **raised** — because light travels from water (denser) to air (rarer) and bends away from the normal, so the bottom appears nearer.
- Letters appear **raised** when viewed through a thick glass slab placed over them.
- A pencil partly immersed in water **appears bent** at the interface — light from the submerged part travels at a different direction than the part above water.
- A lemon kept in water in a glass tumbler appears **bigger** than its actual size when viewed from the sides — due to refraction.

Refraction through a Rectangular Glass Slab



LATERAL DISPLACEMENT

Lateral Displacement: When light passes through a glass slab with parallel faces, the emergent ray is parallel to the incident ray but **shifted sideways**. This perpendicular shift is called lateral displacement.

- At surface AB (air → glass): light goes from rarer to denser → bends **towards** the normal.
- At surface CD (glass → air): light goes from denser to rarer → bends **away** from the normal.
- Since both surfaces are parallel, the bending at AB and CD is equal and opposite → emergent ray is **parallel to incident ray** but laterally displaced.

The Refractive Index**DEFINITION**

Absolute Refractive Index: The Absolute Refractive Index (often simply called the refractive index) of a medium is defined as the ratio of the speed of light in a vacuum to the speed of light in that specific medium.

Refractive Index: A physical quantity that describes how much light slows down and bends when entering a medium. It is the ratio of the speed of light in the first medium to the speed of light in the second medium.

Absolute Refractive Index

$$n = \frac{\text{Speed of light in air (c)}}{\text{Speed of light in medium (v)}} = \frac{c}{v}$$

Refractive Index of Medium 2 with respect to Medium 1

$${}_1n_2 = \frac{\text{Speed of light in medium 1 (v}_1\text{)}}{\text{Speed of light in medium 2 (v}_2\text{)}} = \frac{n_2}{n_1}$$

$${}_1n_2 \times {}_2n_1 = 1 \Rightarrow {}_1n_2 = \frac{1}{{}_2n_1}$$

- Light travels fastest in vacuum with speed $c = 3 \times 10^8 \text{ m s}^{-1}$. In air, the speed is only marginally less.
- Light **slows down considerably** in glass or water.
- **Optically denser medium:** The medium with the higher refractive index. It is **not the same as mass density**. Example: Kerosene ($n = 1.44$) is optically denser than water ($n = 1.33$) but has lower mass density.

Absolute Refractive Index of Some Materials

Medium	Refractive Index (n)	Medium	Refractive Index (n)
Air	1.0003	Crown Glass	1.52
Ice	1.31	Rock Salt	1.54
Water	1.33	Carbon Disulphide	1.63
Alcohol	1.36	Dense Flint Glass	1.65
Kerosene	1.44	Ruby	1.71
Fused Quartz	1.46	Sapphire	1.77
Canada Balsam	1.53	Diamond	2.42

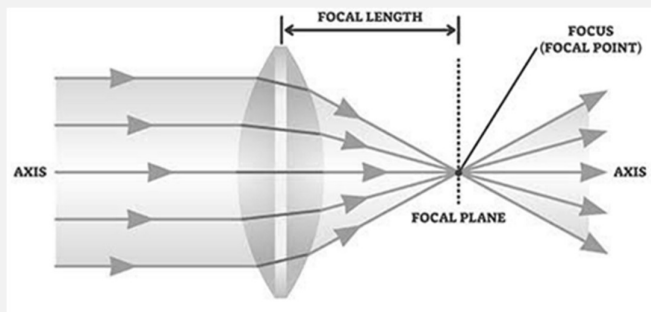
Refraction by Spherical Lenses

DEFINITION

Lens: A transparent material bound by two surfaces, of which one or both are spherical. It refracts light to form images.

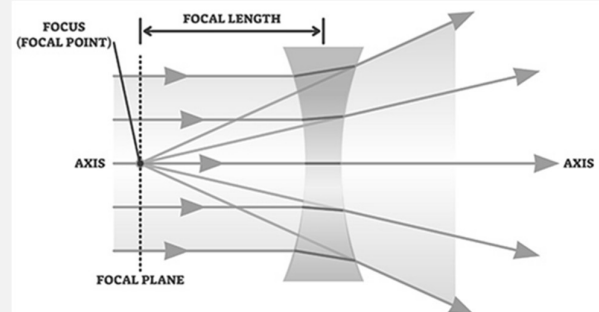
Convex Lens (Converging)

- Bounded by two outward-bulging spherical surfaces
- Thicker at the middle, thinner at the edges
- Converges parallel rays to principal focus F_2
- Focal length is positive (+ve)
- Also called: converging lens



Concave Lens (Diverging)

- Bounded by two inward-curving spherical surfaces
- Thinner at the middle, thicker at the edges
- Diverges parallel rays
- Focal length is negative (-ve)
- Also called: diverging lens

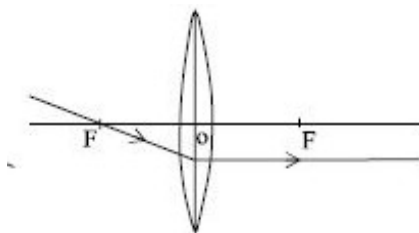


Important Terms – Lenses

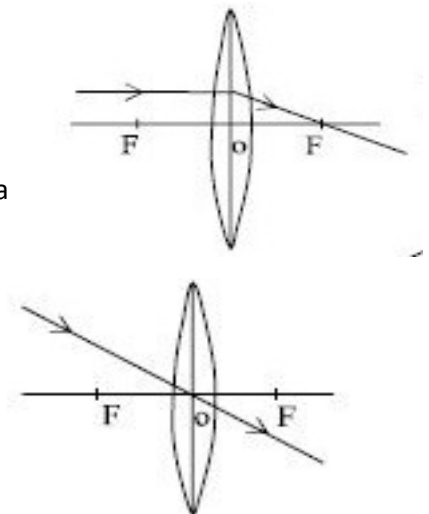
- **Centres of Curvature (C_1, C_2):** The centres of the two spheres of which the lens surfaces form parts.
- **Principal Axis:** The imaginary straight line passing through the two centres of curvature C_1 and C_2 .
- **Optical Centre (O):** The central point of the lens. A ray of light through the optical centre passes without suffering any deviation.
- **Principal Foci (F_1, F_2):** A lens has two principal foci, one on each side. F_1 is the first focal point (object-side), F_2 is the second focal point (image-side).
- **Focal Length (f):** The distance between the optical centre O and the principal focus F.
- **Aperture:** The effective diameter of the circular outline of the lens.

Rules for Drawing Ray Diagrams – Lenses

Rule 1: A ray parallel to the principal axis, after refraction from a convex lens, passes through the second principal focus F_2 . For a concave lens, it appears to diverge from the first principal focus F_1 .



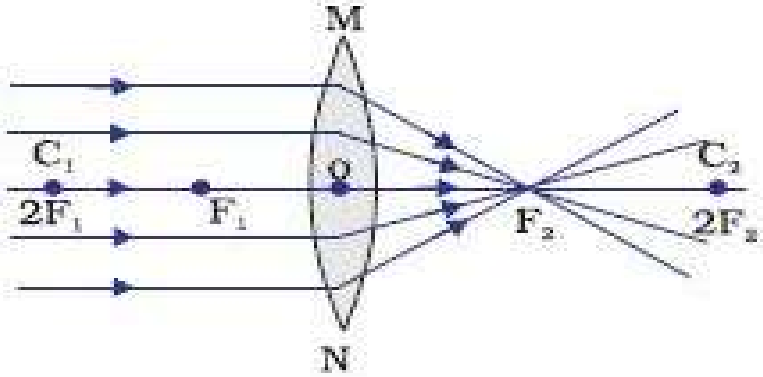
Rule 2: A ray passing through the first principal focus F_1 of a convex lens (or a ray directed towards F_1 of a concave lens), after refraction, emerges parallel to the principal axis.



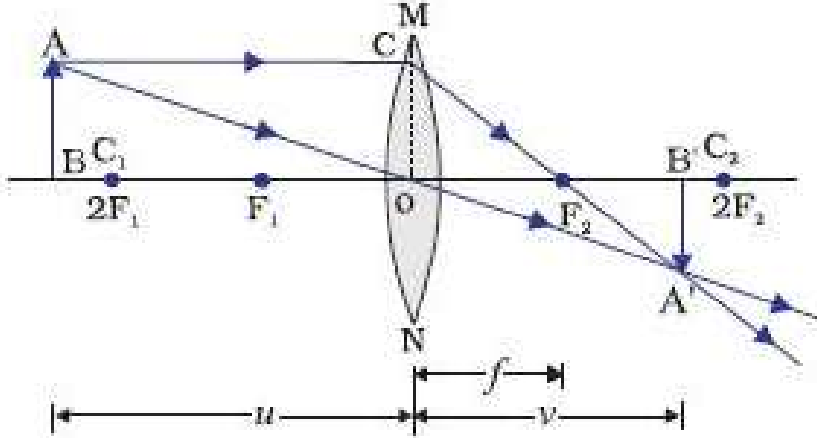
Rule 3: A ray passing through the optical centre O of a lens emerges without suffering any deviation, for both convex and concave lenses.

Image Formation by Convex Lens

Case 1: Object at Infinity

Ray Diagram	
Object Position	At Infinity — object is placed infinitely far from the lens
Object Distance (u)	$u \rightarrow -\infty$ (tends to negative infinity)
Image Location	At Principal Focus F_2 — on the far side of the lens, on the axis
Nature of Image	Real and inverted
Size of Image	Highly diminished — almost a point
Magnification (m)	$m \rightarrow 0$ (tends to zero from negative side)

Case 2: Object Beyond $2F_1$

Ray Diagram	
Object Position	Beyond $2F_1$ — object is placed farther from lens than twice the focal length
Object Distance (u)	$u < -2f$ (magnitude greater than $2f$)
Image Location	Between F_2 and $2F_2$ — on the far side of the lens
Nature of Image	Real and inverted
Size of Image	Diminished — smaller than the object
Magnification (m)	$0 < m < 1$ (m is negative; magnitude less than 1)

Case 3: Object at $2F_1$

Ray Diagram	
Object Position	At $2F_1$ — object is placed exactly at twice the focal length from the lens
Object Distance (u)	$u = -2f$ (magnitude equals $2f$)
Image Location	At $2F_2$ — at twice the focal length on the far side of the lens
Nature of Image	Real and inverted
Size of Image	Same size as the object — neither magnified nor diminished
Magnification (m)	$m = -1$ (magnitude exactly 1, negative sign means inverted)

Case 4: Object Between F_1 and $2F_1$

Ray Diagram	
Object Position	Between F_1 and $2F_1$ — object is placed between the focus and twice the focal length
Object Distance (u)	$-2f < u < -f$ (magnitude between f and $2f$)
Image Location	Beyond $2F_2$ — farther than twice the focal length, on the far side of the lens
Nature of Image	Real and inverted
Size of Image	Enlarged — larger than the object
Magnification (m)	$ m > 1$ (m is negative; magnitude greater than 1 or say $m < -1$)

Case 5: Object at F_1

Ray Diagram	
Object Position	At F_1 — object is placed exactly at the principal focus on the object side of the lens
Object Distance (u)	$u = -f$ (magnitude equals f)
Image Location	At Infinity — refracted rays emerge parallel and never converge to form an image
Nature of Image	Real and Inverted (at infinity — not practically obtainable on a screen)
Size of Image	Infinitely large — size tends to infinity
Magnification (m)	$m \rightarrow -\infty$ (tends to negative infinity)

Case 6: Object Between F_1 and O

Ray Diagram	
Object Position	Between F_1 and O — object is placed between the focus and the optical centre (close to lens)
Object Distance (u)	$0 > u > -f$ (magnitude less than f)
Image Location	Same side as the object — virtual image in front of the lens
Nature of Image	Virtual and Erect (cannot be caught on a screen)
Size of Image	Enlarged — larger than the object
Magnification (m)	$m > +1$ (positive = erect; magnitude greater than 1 = enlarged)

Position of Object	Position of Image	Size	Nature
At infinity	At F_2	Highly diminished, point-sized	Real, Inverted
Beyond $2F_1$	Between F_2 and $2F_2$	Diminished	Real, Inverted
At $2F_1$	At $2F_2$	Same size	Real, Inverted
Between F_1 and $2F_1$	Beyond $2F_2$	Enlarged	Real, Inverted
At focus F_1	At infinity	—	Image not formed
Between F_1 and O	Same side as object	Enlarged	Virtual, Erect

Image Formation by Concave Lens

Case 1: Object at Infinity

Ray Diagram	
Object Position	At Infinity — object is placed infinitely far from the concave lens
Object Distance (u)	$u \rightarrow -\infty$ (tends to negative infinity)
Image Location	At virtual Focus F_1 — on the same side as the object
Nature of Image	Virtual and Erect (cannot be caught on a screen)
Size of Image	Highly diminished — almost a point
Magnification (m)	$m \rightarrow 0+$ (positive, approaching zero)

Case 2: Object at Any Finite Distance

Ray Diagram	
Object Position	Between infinity and O — object placed at any finite distance from the lens
Object Distance (u)	Any value: $0 > u > -\infty$ (any negative value)
Image Location	Between F_1 and O, same side as the object; closer to lens as object moves closer
Nature of Image	Virtual and Erect (cannot be caught on a screen)
Size of Image	Diminished — always smaller than the object
Magnification (m)	$0 < m < 1$ (positive = erect; magnitude less than 1 = diminished)

Position of Object	Position of Image	Size	Nature
At infinity	At focus F_1 (same side)	Highly diminished, point-sized	Virtual, Erect
Between ∞ and O	Between F_1 and O	Diminished	Virtual, Erect

★ Remember

A concave lens ALWAYS produces a virtual, erect, and diminished image irrespective of the position of the object.

Lens Formula and Magnification

LENS FORMULA

Gives the relationship between object distance (u), image distance (v), and focal length (f) for a spherical lens.

Lens Formula

$$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$$

Lens Magnification (m)

$$m = \frac{h'}{h} = \frac{v}{u}$$

SIGN CONVENTION FOR LENSES

- Measured from the **optical centre O**
- All rules same as for mirrors (right = +ve, left = -ve, up = +ve, down = -ve)
- Focal length of **convex lens is positive (+ve)**; Focal length of **concave lens is negative (-ve)**

Power of a Lens

DEFINITION

Power of a Lens (P): The degree of convergence or divergence of light rays achieved by a lens. It is defined as the reciprocal of the focal length of the lens.

$$P = \frac{1}{f \text{ (in metres)}} \quad \text{Unit: Diopetre (D)} \quad 1D = 1 \text{ m}^{-1}$$

KEY POINTS – POWER

- 1 Diopetre:** The power of a lens whose focal length is 1 metre.
- Power of **convex lens** → **positive (+ve)**; Power of **concave lens** → **negative (-ve)**.
- Combined power of lenses in contact:** $P = P_1 + P_2 + P_3 + \dots$
- Opticians prescribe lenses with their power (e.g., +2.0 D = convex, $f = 0.5 \text{ m}$; -2.5 D = concave, $f = -0.40 \text{ m}$).

Key Experimental Observations

Observation 1 – Curved Spoon as Mirror

- The concave side of a shining spoon acts as a **concave mirror**.
- When held close: a magnified, erect, virtual image of your face is seen (object between P and F).
- As the spoon is moved away: the image inverts, diminishes, becomes real.
- The convex side acts as a **convex mirror** — always gives a diminished, erect, virtual image.

Observation 2 – Concave Mirror Focus

- Hold a concave mirror facing the Sun. Direct the reflected light on a sheet of paper.
- Move the paper back and forth until a bright sharp spot appears — this is the **focus F**, and the distance from mirror to paper is the **focal length**.
- The paper may begin to burn since the Sun's rays, being parallel, are converged at F, generating intense heat.

Observation 3 – Coin in Water (Refraction)

- A coin placed at the bottom of a bowl and just out of sight reappears visible when water is poured in.
- Light from the coin refracts at the water surface (going from denser to rarer medium), bending **away from the normal**.
- This makes the coin appear **raised** from its actual position — an effect of refraction.

Observation 4 – Glass Slab Lateral Displacement

- A straight line drawn on paper, viewed through a glass slab placed at an angle, appears **bent or shifted** at the edges of the slab.
- Viewed from the top, the line appears **raised** — because light refracts and slows down in glass.
- The emergent ray is parallel to the incident ray but laterally shifted — this is called **lateral displacement**.

Observation 5 – Convex Lens Burning Paper (Focal Length)

- Hold a convex lens facing the Sun. Focus the light on paper — a bright sharp spot forms.
- The bright spot is the **real image of the Sun** at focus F_2 .
- The paper catches fire because the Sun's parallel rays converge at one point, concentrating heat.
- Distance from lens to paper = **approximate focal length of the lens**.

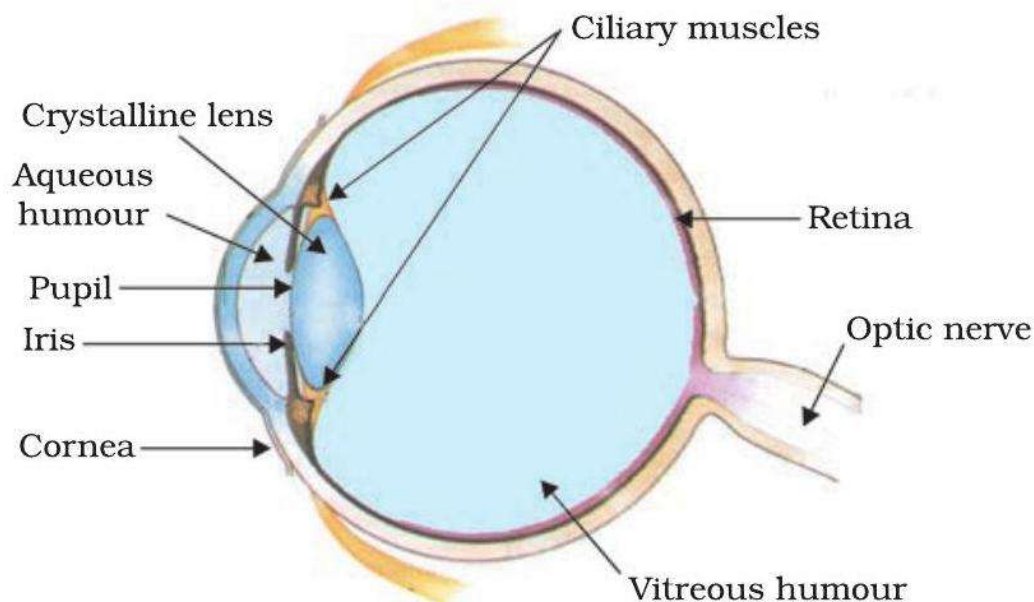
THE HUMAN EYE AND THE COLOURFUL WORLD

The Human Eye

DEFINITION

Human Eye: A natural optical instrument that functions like a camera. Its lens system forms an image on a light-sensitive screen called the retina, enabling us to see the world.

Parts of the Human Eye and Their Functions



- **Cornea:** A thin transparent membrane that forms the transparent bulge on the front surface of the eyeball. **Most of the refraction** of light entering the eye occurs at the cornea.
- **Iris:** A dark muscular diaphragm that lies behind the cornea. It controls the **size of the pupil** and thus regulates the amount of light entering the eye.
- **Pupil:** The small opening in the iris. It expands in dim light and contracts in bright light to regulate the **amount of light** entering the eye.
- **Crystalline Lens:** A fibrous, jelly-like transparent structure. Its curvature is changed by ciliary muscles to fine-adjust the **focal length** to focus on objects at different distances.
- **Ciliary Muscles:** Muscles that control the **curvature of the crystalline lens**, thereby adjusting its focal length for accommodation.
- **Retina:** A delicate membrane at the back of the eye having an enormous number of light-sensitive cells. When illuminated, these cells generate **electrical signals** sent to the brain via the optic nerve.
- **Optic Nerve:** Carries the electrical signals from the retina to the **brain**, which interprets these signals and creates the perception of the image.
- **Aqueous Humour:** A watery fluid between the cornea and the crystalline lens. Helps maintain pressure and shape of the front of the eyeball.
- **Vitreous Humour:** A jelly-like substance filling the space between the crystalline lens and the retina. Maintains the **spherical shape** of the eyeball.

★ Remember

The eyeball is approximately spherical with a diameter of about **2.3 cm**. Most refraction occurs at the cornea; the lens only provides fine adjustment.

Power of Accommodation

DEFINITION

Power of Accommodation: The ability of the eye lens to adjust its focal length to focus objects clearly at different distances on the retina. This is achieved by changing the curvature of the crystalline lens through the ciliary muscles.

- **Looking at distant objects:** Ciliary muscles **relax** → lens becomes **thin** → focal length **increases** → parallel rays from distant objects focus on retina.
- **Looking at nearby objects:** Ciliary muscles **contract** → lens becomes **thick** → focal length **decreases** → diverging rays from close objects focus on retina.

NEAR POINT AND FAR POINT

- **Near Point (Least Distance of Distinct Vision):** The **minimum distance** at which the eye can see objects clearly without strain. For a young adult with normal vision: **25 cm**.
- **Far Point:** The **farthest distance** up to which the eye can see objects clearly. For a normal eye: **Infinity**.
- A normal eye can see objects clearly that are between **25 cm and infinity**.

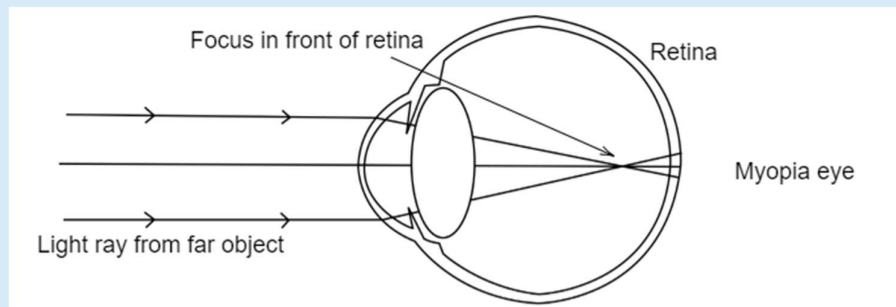
Defects of Vision and Their Correction

The eye may gradually lose its power of accommodation. The vision becomes blurred due to refractive defects. The three main defects are: **(i) Myopia (ii) Hypermetropia (iii) Presbyopia and (iv) Cataract**. All are correctable with appropriate spherical lenses.

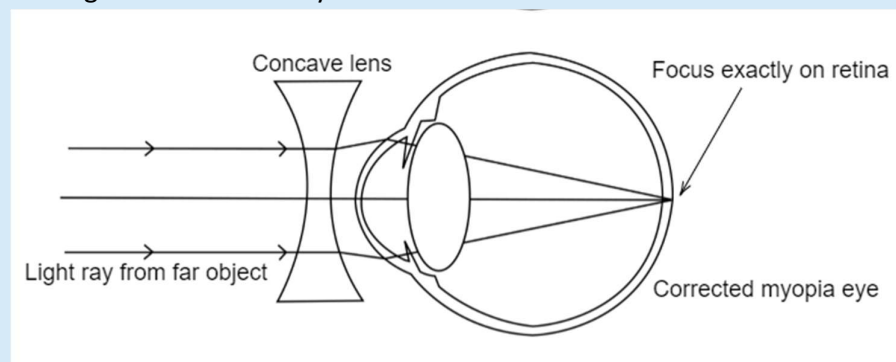
(a) Myopia – Near-Sightedness

MYOPIA

- **Also known as:** Near-sightedness.
- **Can see:** Nearby objects clearly; cannot see distant objects distinctly.
- **Far Point:** Nearer than infinity (only a few metres away).
- **Defect:** Image of a distant object is formed **in front of the retina** instead of on it.
- **Causes:** (i) Excessive curvature of the eye lens, or (ii) Elongation of the eyeball.



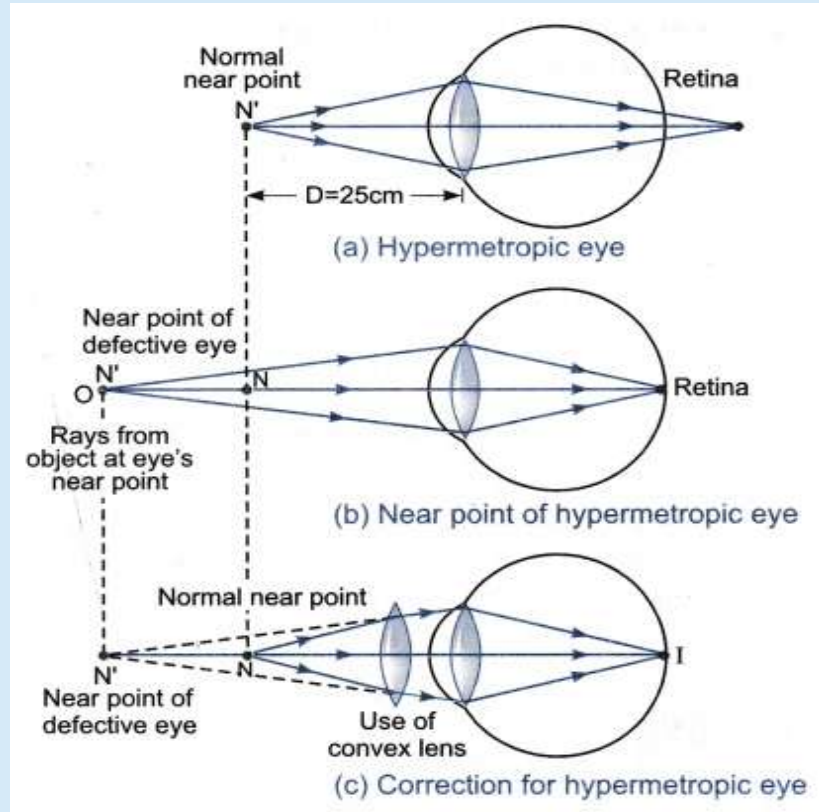
- **Correction:** A **concave lens** of suitable power is used. It diverges the light rays before they enter the eye so that the image is formed exactly on the retina.



(b) Hypermetropia – Far-Sightedness

HYPERMETROPIA

- **Also known as:** Far-sightedness.
- **Can see:** Distant objects clearly; cannot see nearby objects distinctly.
- **Near Point:** Much farther than the normal near point (25 cm).
- **Defect:** Light rays from a close by object are focused **behind the retina** instead of on it.
- **Causes:** (i) Focal length of the eye lens is too long (eye lens is too flat), or (ii) Eyeball has become too small.



- **Correction:** A **convex lens** of appropriate power is used. It provides the additional converging power required so that the image is formed on the retina.

(c) Presbyopia – Age-Related Vision Defect

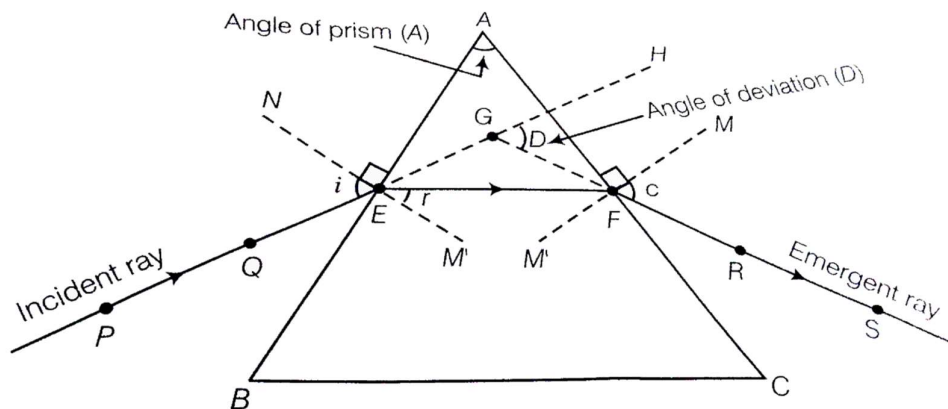
PRESBYOPIA

- **Definition:** The gradual loss of the power of accommodation of the eye that usually occurs with ageing.
- **Symptoms:** The near point gradually recedes further away; difficulty in seeing nearby objects comfortably and distinctly without corrective glasses.
- **Causes:** (i) Gradual weakening of the ciliary muscles, and (ii) Diminishing flexibility of the crystalline lens with ageing.
- **Special case:** A person may suffer from both myopia and hypermetropia simultaneously.
- **Correction: Bi-focal lenses.** The upper portion contains a **concave lens** (for seeing distant objects clearly) and the lower portion contains a **convex lens** (for reading and near vision).
- **Modern corrections:** Contact lenses or surgical interventions (such as LASIK) can also correct all refractive defects.

(d) Cataract – Age-Related Vision Defect

Cataract: A condition in which the crystalline lens of the eye becomes **milky and cloudy** at old age. This causes partial or complete loss of vision. Vision can be restored through **cataract surgery** in which the lens is replaced.

Refraction of Light through a Prism



ANGLE OF DEVIATION

Angle of Deviation ($\angle D$): The angle between the direction of the incident ray and the direction of the emergent ray. The prism bends the emergent ray relative to the incident ray.

- A triangular glass prism has two triangular bases and three rectangular lateral surfaces inclined to each other.
- Angle of Prism ($\angle A$):** The angle between the two lateral refracting faces of the prism.
- At face AB (air $\rightarrow$ glass): light bends **towards** the normal (entering denser medium).
- At face AC (glass $\rightarrow$ air): light bends **away** from the normal (entering rarer medium).
- The net effect is that the emergent ray FS is **NOT parallel** to the incident ray PE — unlike in a glass slab.
- Notation: **PE** = Incident ray; **EF** = Refracted ray inside; **FS** = Emergent ray; $\angle i$ = angle of incidence; $\angle r$ = angle of refraction; $\angle e$ = angle of emergence.

★ Remember

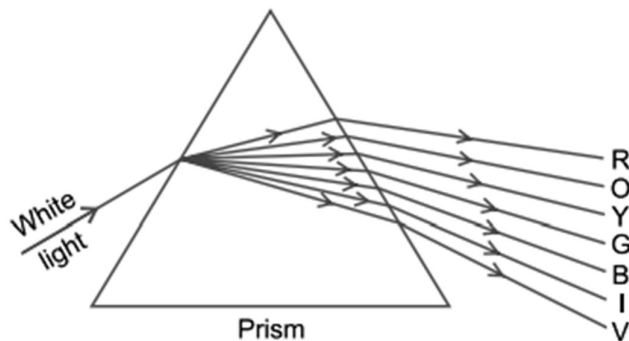
The light used in above diagram is Monochromatic light, which has only one colour.

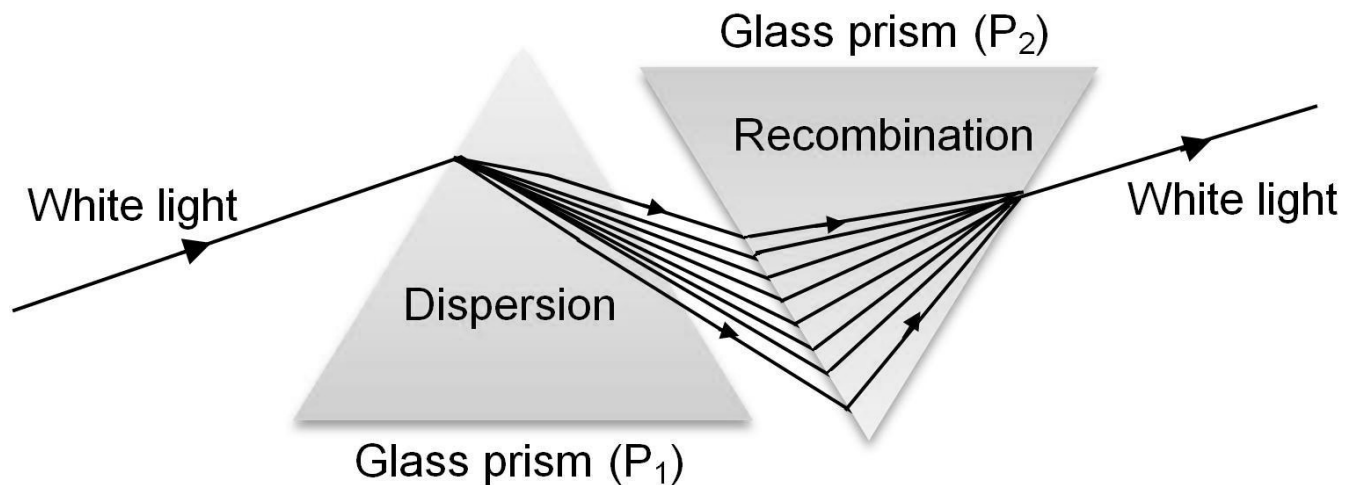
Dispersion of White Light by a Glass Prism

DEFINITION

Dispersion: The splitting of white light into its constituent seven colours (spectrum) when it passes through a prism. This occurs because different colours of light have different refractive indices in the prism material and hence bend through different angles.

- The band of seven coloured components of white light is called the spectrum.
- Order of colours (acronym **VIBGYOR**):
Violet – Indigo – Blue – Green – Yellow – Orange – Red
- Red** light bends the **least** (has the longest wavelength, ~ 700 nm).
- Violet** light bends the **most** (has the shortest wavelength, ~ 400 nm).
- Each colour has a different refractive index in glass $\rightarrow$ each colour emerges along a different path $\rightarrow$ they spread out into a spectrum.
- Dispersion occurs in a prism because its non-parallel surfaces bend light rays in the same direction, spreading them further apart. In a glass slab, opposite faces are parallel. The second face cancels out the dispersion caused by the first, recombining the colours to form white light.



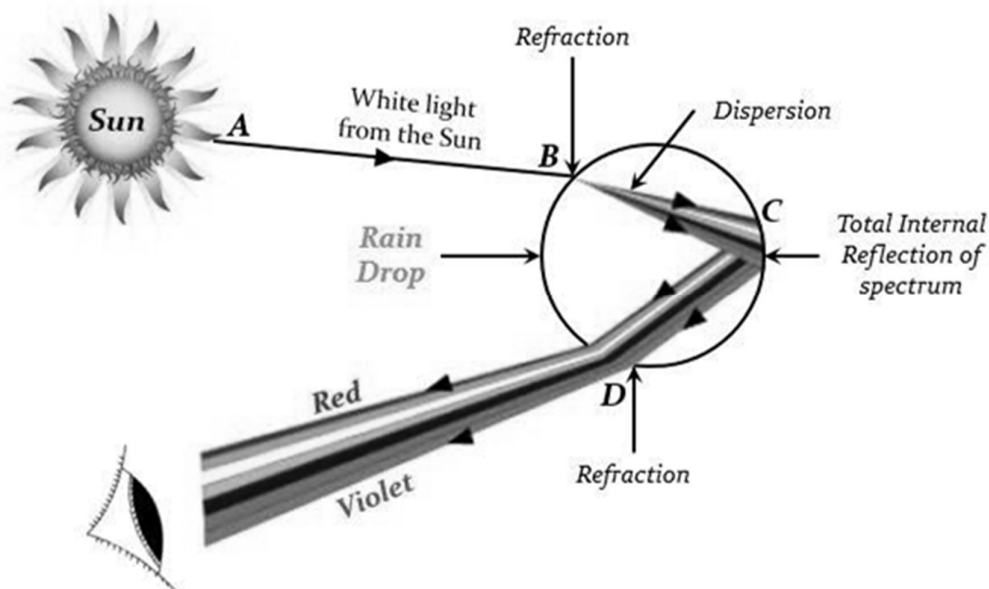


- Isaac Newton was the first to use a glass prism to obtain the spectrum of sunlight.
- He used a second prism to split the colours further — but got no new colours, proving the colours are fundamental.
- He then placed a second identical prism **inverted** with respect to the first — all colours passed through and recombined to give **white light**.
- This proved that **white light (sunlight) is made up of seven colours**
- Any light giving a spectrum similar to sunlight is called **white light**.

Rainbow

RAINBOW

Rainbow: A natural spectrum appearing in the sky after a rain shower. It is caused by the **dispersion of sunlight by tiny water droplets** present in the atmosphere.



- Each water droplet acts as a tiny prism.
- Inside each droplet, light undergoes: **Refraction → Internal Reflection → Refraction**.
- A rainbow is always formed in the direction **opposite to the Sun**. The Sun must be behind the observer.
- Different colours reach the observer's eye from different droplets at different angles due to dispersion.
- A rainbow can also be seen through a **waterfall or water fountain** on a sunny day with the Sun behind you.

Atmospheric Refraction

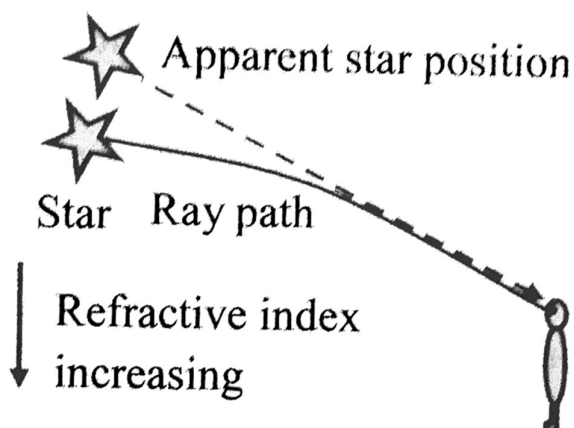
DEFINITION

Atmospheric Refraction: The refraction of light by the Earth's atmosphere. The atmosphere has different layers with gradually changing refractive index (density). Light passing through it continuously changes direction.

Objects seen through hot turbulent air (above a fire or radiator) appear to waver — the hotter air is lighter (less dense) and has a slightly lower refractive index. The changing physical conditions cause the apparent position of the object to fluctuate. This is atmospheric refraction on a small scale.

Twinkling of Stars**TWINKLING OF STARS**

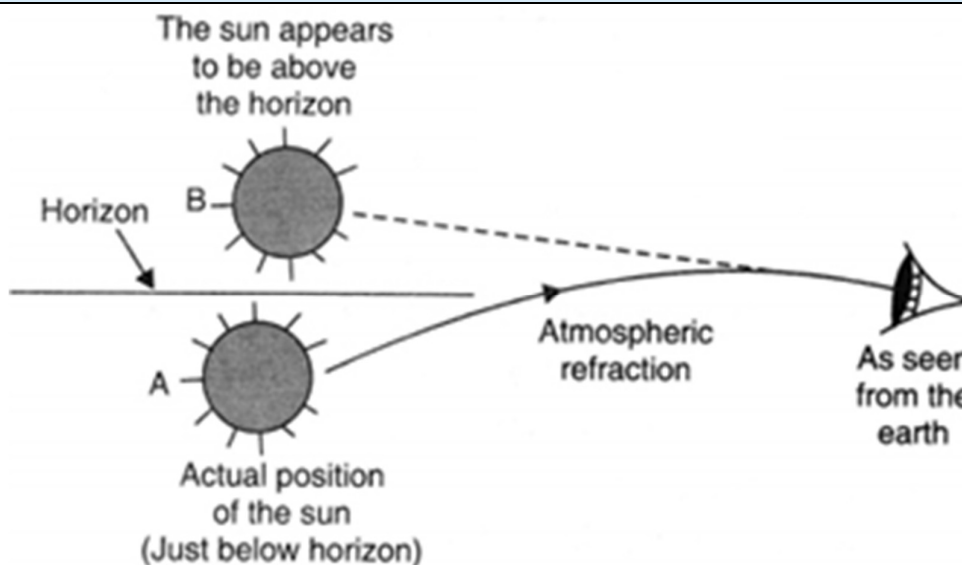
Why stars twinkle: As starlight travels through the atmosphere, it undergoes continuous refraction. The atmosphere has a medium of gradually changing refractive index. The apparent position of the star keeps fluctuating since atmospheric conditions are not stationary. Stars are very distant, so they appear as **point sources** of light. As the path of rays keeps varying, the amount of starlight entering the eye flickers — the star appears sometimes brighter, sometimes fainter — this is the **twinkling effect**.



Why planets do not twinkle: Planets are much closer to the Earth and therefore appear as **extended sources** (not point sources). If we consider a planet as a collection of many point sources, the total variation in light from all the individual point sources averages out to zero — thus nullifying the twinkling effect.

Advance Sunrise and Delayed Sunset.**ADVANCE SUNRISE AND DELAYED SUNSET**

The Sun is visible about **2 minutes before** the actual sunrise and about **2 minutes after** the actual sunset because of atmospheric refraction. Light from the Sun near the horizon travels through a greater thickness of atmosphere and is continuously refracted (bent towards the normal as it enters denser layers), making the Sun appear above the horizon even before it has physically risen.



- By **actual sunrise**, we mean the actual crossing of the horizon by the Sun.
- The **apparent flattening** of the Sun's disc at sunrise and sunset is also due to atmospheric refraction (the lower edge of the Sun refracts more than the upper edge).

Scattering of Light

DEFINITION

Scattering of Light: When a beam of light encounters fine particles (much smaller than or comparable to the wavelength of light), the particles absorb the incident light and re-emit it in all directions. This is called scattering. The interplay of light scattering with atmospheric particles gives rise to several natural phenomena.

10.6.1 Tyndall Effect

DEFINITION

Tyndall Effect: The phenomenon of scattering of light by colloidal particles (or fine particles of dust and smoke) in a medium, making the path of a beam of light visible. Named after physicist John Tyndall.

- The Earth's atmosphere is a heterogeneous mixture — it contains smoke, tiny water droplets, suspended dust particles, and air molecules.
- When a beam of light strikes such fine particles, it is scattered in all directions — the path of the beam becomes **visible**.
- **Colour of scattered light depends on the size of particles:**
- Very fine particles (smaller than λ of light) → scatter mainly **blue light** (shorter wavelengths scattered more strongly).
- Larger particles → scatter light of **longer wavelengths**. Very large particles → scattered light may appear **white**.
- **Examples:**
- A fine beam of sunlight entering a **smoke-filled room** through a small hole — the beam path is visible due to Tyndall Effect.
- Sunlight passing through the **canopy of a dense forest** — tiny water droplets in mist scatter the light, making beams visible.

10.6.2 Why is the Colour of the Clear Sky Blue?

Blue Sky Explanation: Molecules of air and fine particles in the atmosphere are very small — smaller than the wavelength of visible light. These particles are much more effective in scattering light of **shorter wavelengths** (blue end) than longer wavelengths (red end). When sunlight passes through the atmosphere, fine particles scatter the **blue colour more strongly** than red. This scattered blue light enters our eyes from all directions, making the sky appear **blue**.

- Red light has a wavelength about **1.8 times** greater than blue light → scattered far less.
- If the Earth had no atmosphere, there would be no scattering → the sky would appear **dark/black**

Why Danger Signal Lights are Red

REASON

Red light has the **longest wavelength** in the visible spectrum. It is **least scattered** by fog, smoke, and dust. Therefore, red light can travel the longest distance through fog or smoke and still be seen in its original colour. This is why danger signals, stop lights, and tail lights are red in colour.

Why the Sun Appears Red at Sunrise and Sunset

At sunrise and sunset, the Sun is near the horizon. Sunlight has to travel through a **much greater thickness of atmosphere** compared to midday. Most of the **blue and shorter wavelengths are scattered away** along the path. Only longer wavelengths — **red and orange** — reach our eyes. Therefore, the Sun appears reddish-orange at sunrise and sunset. The sky around it also takes on orange and red hues.